BGIN Block#11 Meeting Report

- BGIN is a multi-stakeholder discussion body / distributed think tank.

Wallet Governance

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BGIN Block#11

Wallet Governance

Date: Monday, October 22, 2024

Location: Washington DC, United States

Duration: Approx. 90 minutes

Summary

Biometric Key Management

Question 1:
What are biometric schemas, and is it possible to read them?

Answer:
Biometric schemas are specific files designed for secure storage of biometric data. The technology to read these schemas is typically proprietary, making it challenging to access or use them outside of their intended systems. Even if the schema data were leaked, it would be difficult to interpret or use effectively without the proprietary technology.

Question 2:
How is a biometric different from data like a Social Security number?

Answer:
Unlike data such as a Social Security number, which is standalone, biometric data relies on physical traits like fingerprints. Even if the biometric data were leaked, linking it back to an individual is more complex, adding a layer of security.

Question 3:
Does using biometric security decrease the convenience of wallet use?

Answer:
Yes, using biometric data can reduce convenience, as users must go through additional verification steps. However, biometric-bound wallets do not store the biometric data directly but rather derive public information from each interaction, which is not stored anywhere. This setup reduces privacy risks but can impact usability.

Question 4:
Could biometric authentication alone achieve complete security?

Answer:
Biometric authentication cannot completely eliminate error rates, so alternative solutions like multi-factor authentication should be considered for critical systems. Relying solely on biometrics may not be sufficient due to the inherent limitations in biometric data accuracy.

Question 5:
Do you use only fingerprint data to generate the signing key, or is additional information required?

Answer:
This depends on the biometric signature scheme in use. Some systems, like Hitachi's, use only fingerprint data, while others might incorporate additional "helper data." This helper data does not contain confidential information and cannot reconstruct the original biometric information, serving only to enhance security.

Question 6:
What about the potential risks of losing biometric data or parts of the body associated with it?

Answer:
If someone loses the part of the body used for biometric authentication (like a fingerprint), they could lose access to the wallet. Biometric security represents both a strength and a weakness, as it is unique to the individual. Future technologies, including generative AI, may potentially reconstruct biometrics, which would be a security challenge if biometrics are the only form of authentication.

Question 7:
How is entropy managed in biometric-based signature schemes to maintain security?

Answer:
Maintaining high entropy in biometric signatures is essential for security, and although biometric systems have some entropy loss, it is necessary to have alternative systems or multi-factor authentication to reinforce security.

Question 8:
Could biometric-locked wallets be combined with other key management methods?

Answer:
Yes, biometric-locked wallets could support additional key management methods. For instance, a wallet could use biometrics for low-value transactions while requiring multiple authentication factors for higher-value transactions. This tiered approach could balance security and convenience.

Question 9:
How do biometric systems handle minor alterations in biometric data, like slight changes in fingerprints?

Answer:
To account for variations, biometric systems use "helper data" that compensates for minor differences in biometric information. This helper data adjusts for small discrepancies in features, allowing the biometric data to match without needing exact replication.

Question 10:
Is there any need for external storage in biometric-bound wallets to handle data fuzziness?

Answer:
Typically, external storage for keys is unnecessary because biometric-bound wallets incorporate procedures like activation and matching stages. These processes align registration data with real-time biometric data, enabling secure key generation without storing extra data externally.

Accountable Wallet

Question 1:
Will the accountable wallet concept create a duality in the ecosystem, possibly affecting fungibility, like pricing differences between accountable and non-accountable Bitcoin?

Answer:
Yes, introducing accountable wallets could lead to a division in the ecosystem between accountable and non-accountable systems. This has raised concerns about potential differences in the pricing of accountable versus non-accountable tokens, such as Bitcoin. This impact on fungibility was a major topic at the Bitcoin Tokyo event.

Question 2:
How can we build incentives for using accountable wallets?

Answer:
Positive ecosystem inclusion may serve as an incentive, as users often want to participate in trusted environments. This could create a "snowball effect," making accountable wallets essential for accessing certain financial opportunities, thus encouraging their use.

Question 3:
What role could accountable wallets play in decentralized finance (DeFi), specifically in compliant, walled gardens?

Answer:
Accountable wallets could be instrumental in creating compliant, walled gardens within DeFi. By using block filters, transactions from sanctioned ecosystems could be identified, supporting reputation-based incentives, reducing business risks, and benefiting individual investors.

Question 4:
Will accountable wallets ease or increase user responsibility for privacy management?

Answer:
Expanding privacy management responsibilities onto users does add to their burdens. However, accountable wallets could mitigate this by offering incentives to users who take ownership of their data, potentially shifting privacy management to a beneficial, positive-sum outcome where users gain from their data control.

Question 5:
Could accountable wallets benefit validators in addition to Virtual Asset Service Providers (VASPs)?

Answer:
Yes, users and validators could benefit if transactions are validated using accountable wallets. However, there are concerns around centralization, as seen with F2Pool's decision to exclude flagged transactions based on sanctions, which was criticized for its centralized approach. Accountable wallets should ideally support decentralized, individual decisions rather than relying on centralized control.

Question 6:
Is there a way to avoid centralization issues when implementing accountable wallets?

Answer:
One potential solution is creating an association set where accountable wallets only interact with transactions mined by particular relayers compliant with sanctions like OFAC. This decentralized approach could combine accountable wallets with privacy pools and compliance-focused tools, such as NEV (non-extractive value) bots, rather than relying on centralized entities.

Question 7:
Will users voluntarily adopt accountable wallets to avoid association with harmful activities?

Answer:
Voluntary adoption may be challenging, as many users don't prioritize privacy, and avoiding illegal involvement may not be a strong enough motivator. Some believe enforcement may be necessary to ensure widespread adoption, although the concept could gain acceptance over time by providing a whitelist for exclusive opportunities, such as airdrops or new token access.

Question 8:
How complex would it be to implement accountable wallets, and what compliance measures would be needed?

Answer:
Implementing accountable wallets might require extra steps, like signing an additional transaction or querying an on-chain registry for compliance verification. Although this could increase development complexity, these steps could be worthwhile for enhancing user security and preventing negative interactions.

Question 9:
Can an accountable wallet be revoked, and would monitoring be necessary?

Answer:
Revocation does not necessarily require monitoring. Accountability could decay over time, with recent transactions holding higher legitimacy. Wallet holders could refresh their legitimacy

through continued control. If a bad actor takes over, a trusted third party could revoke the wallet's attestation, isolating the issue to that specific instance.

Question 10:
How would the system handle the steady state of an accountable wallet, especially if a bad actor later misuses it?

Answer:
The system would need real-time reputation decay, allowing users to refresh their reputation through control retention. If a bad actor assumes control, a third party could revoke the wallet's token, possibly impacting related attributions and discouraging misuse over time.

Transcript

which you don't see everyone this is all the governance and the other one is internet governance yeah so it's the comparison with the gaming session and this one's the accountable wallet such as. Okay, good, bad. That's good. Thank you. I agree, yeah, it's just supposed to be the most interesting. to share everyone's. So what I've done? Thanks. Absolutely. You can… Okay, it's fine. Any question, maybe this laptop so if you want to share your screen instead of that sounds awesome. I'm just going to, I really need a quick break. Oh, gosh. course. Yeah, I was going to say that. It's the last two months. Sorry. Inclusion. I said, oh, yeah. What's difficult. He was actually really interesting. the idea of preaching on this It's always something much better. a global message. And then there's the link. What does this look like a built-in patient? you're not doing anything with me quarterly And every time you get your paycheck, you've got nothing you are like, I do like YYC. I did a negated system. I get good rates of return. I get all the summary. And other people are like, I don't know about this. There's an argument And I think that specifically which ones the system like the the gated system It's true. Thank you. Thank you. Oh, yes. Hello, hello, everyone. open to the final session of the day to begin block 11 fortunately this session is one close to my heart on wallet governance um typically the world governance study report paper that we've been contributing to and preparing and writing in the identity key management and privacy working group. As well as the accountable wallet research that's been undergone over the last year and a half. Thanks to Ms. Hunter's son. So I'll just quickly introduce the session. And then we'll move to Seppy Sun, who's going to provide us a presentation on wallet governance to provide an overview of the wallet ecosystem I will then sort of proceed to brief the room on the wallet governance paper where we can use support in contributions as well as Kind of like the angle and the questions that we're asking. Because they're pretty important questions and There's a lot of Over the past two days, what I've realized is that there's a lot of interface and touch points with wallets in this entire ecosystem. So having a shared common understanding, which is part of Begin's mission. should be front and center. And I think Willits is

a fantastic place to start. So without further ado So invite Saki-san to present her research on wallet governance. screen? Yeah, please. Before I get to a major, let me introduce myself briefly I'm Sekiotsky. I belong to NAC Corporation in Japan. NEC is strong in biology authentication, especially facial recognition. And I have a background in cryptography and my current research topic is combining biometrics and cryptography. Today, we need to develop key management types of glutenoids. I'm focusing on biogeochemical management. Makes grades. It's skip this, right? Yeah. The purpose of my communication is to improve the key management section of the work governance study report The purpose of the session is made for first we want to achieve a formal understanding of the importance of key management Mr. Gant, we want to give a customer's understanding of where it's with various key management types. And finally, we want to share information about climate change management and lay the foundations for future discussions and i'd like to key opinions from their standpoints. Next slide, please. Thank you so did you want to swap and stand here now that the presentation's here? Thank you. Let's review the Laurel keys in Brooklyn White. A brochure work is a tool for managing and trading cryptocurrency or nfts and digital signatures are used to generate value transactions A word account is linked to our key pair consisting of a signing key, which is a private key, and a verification key, which is a public key The verification key functions as a word address and you can think of it like account number On the other hand, the Sanyuki functions as a warrant The signing care functions like a password and it is used to sign the contents of the given transactions In other words, access rights to means having a rights to trans assets. So. For this reason, the management of Sanya piece is important for blockchain Worth. On speaking awards, this includes the generate and stretch and management of various things other than san yokes. But this section will focus on the early management of Sanyuki's. So what specific requirements are there for planning care management? And we have identified three main points to consider. these security vulnerability and convenience. In terms of security. The key question is whether or not there is a risk for the unauthorized access of signing keys. We'll consider the various factors listed on that's right. And in terms of availability. The key question is without note the user can access the financial key even in the event of a failure These two are particularly important in terms of access rights to assets, but when it comes to practical use for customers. Convenience is also important. with most products, key management is something that many developers think about When designing the product. And users are often our rare orbit. But Ms. Brookecha, worried users meet to around key management in order to protect their own assets. Okay. But in general, it is difficult to achieve the balance of security, availability, and convenience of key management. So customers should choose the appropriate appropriate key management methods according to use case or desired functions or their preference Next, we will move on the explanation of the word types listed in the report. There are various ways to classify words, and this will surmise these h types. the classification of custodial and non-custodial is based on the key management entity. And the classification of software and hardware wallets is based on the location of the key And most signature words and multi-party computational words, MCC worries. the words that use redundant secret information this redundant part improves security

and availability compared to single keyword And small contract were it that automate transactions using programs on Ethereum. These seven types are probably most familiar to everyone here. So for more details, please refer to the report. And we are now exploring the biometric world defined by the IKP working group. This is a type that uses pharmacological information for key management. where this concept has not been widely used, we think that this type of word will become established in the future. Now let's talk about this biometric words. In the work of mastery reports. We define biometric words. As a general term for words, they use biomedical information to human management. Within this, we define two types of wallets biometric blocked and biometric bound for it The log type uses biometric information to activate the key. And this includes Bamajig. This includes hardware wallets, which uses spamtrick authentication to activate the keys such as decent biometric wallet And other example is like a phytotype right And the biomagic bias type word use summative information as a substitute for the typhanguy. Although it is not yet widespread, various technologies make this possible. Such as further commitments or fuzzy signatures that are being researched and recently and hashport announced that the start of a collaboration to implement a biometric bound web3 wallets The motivation for biometric management is nearly identical to your explanatory information for biometric authentication. This is because biometric information has strong binding with its owner. And it is easy to use. Specifically, this is because 500 information has the following three properties. we never received and no one can steer it and we can lend it to someone else In this slide, we summarize features common to post flavjic words. first grades to security. Obviously, it really depends on the accuracy of packaging concentration. And in practice, there is a list of impersonation using copied photos or videos. So pregnant. That's true. Preventing this requires system level enhancements. such as presentation or target detection or digital injection mitigation and ai based thread addictions. And in terms of availability. It damages information on earth is mandatory, there is a single point of failure. For example, there's a possibility that biometric information cannot be used or does not function This can be due to the refund. environmental factors or injury? And in terms of convenience, it is highly convenient and this should not need explanation On the other hand, there are concerns about privacy. And because biometric information is difficult to change. So it is necessary to consider where to extract and store biometry information. And in what state to store it in? For example. states include photos or future big stories and impression of magic future big stores. Up to this point, we have talked about the common features of biometric words And next, we explain the difference between biometric clock and biometric bond types. The main difference between the two types are the strengths of the binding between the key and its owner. An example of a lot type wallets on the shown on the left And an example of a bound where it is shown on the right. We were starved with an explanation of the log type. The user presses their fingerprints on the device And if the authentication is successful. The store key on the device gets activated And this kid then used to generate a signature While activating the key becomes easier. storing the key on the device poses risks. there includes the likelihood of hacking and the potential roles of the device So in this context, the binding can be considered weak. Next, we're explaining the biometric balance type

the user inputs their fingerprints information into the device. And signature is denoted from it. So no confidential information is stored So it is hard to forge job is that users biometric information. In this sense, it can be said that the binding is strong the binding is strong Okay. Also, there are differences in the strengths of the binding between the key and the owner industry, the degree to which the key is used to identify user differs depending on the use case. So it is necessary to make a choice accordingly. This is all for my presentation. So any comments or questions?

So there are different biometric schemas. So they're collections of It is a file, but it's very specific schema and um If I'm answering this correctly, but I know what you mean and I think that if is possible to read that, but the technology in order to read the biometric schema is set up a proprietary And it's difficult to get your hands on kind of thing um but yeah We had that question actually two blocks ago and Blair from doltry provided a really good answer on the different types of biometric schemas that are out there. And I would say that the one that is used for this would be almost like proprietary to the wallet In particular, that is using that biometric um like c and show So even if it was leaked, it's going to be very hard for it to be relevant or usable by anyone else. Yep, and also relies on The difference between your social security number is that it is data in and of itself. a biometric relies on like a fingerprint, right? So even if that data set leaked to connect that biometric to you as a person is another thing altogether. under eye. to add some comments on your first and second question. But first. a question uh i also recognize that um drinks, Leibnish check. decrease the convenience of biometric warriors. So, yeah. So yeah, I recognize it. And for the second question. maybe the biometric bond war is while incarceration because it Because you cannot need store your biometric information anywhere in this type of words. And just each time you Signing. you extract your public information. and um So it is not stored anywhere. Commons for you christians. I totally agree with the Phineas or the it's a very important point that we we need to face as kind of illustrations Based on the biometry Thank you. But in the context of security. We can't reduce this kind of the private ratio so that error ratio to zero So the slide so we need to have a kind of a uh alternative solution like multi-factor authentication software So we should depend on the kind of technologies but we shouldn't depend on that only ethnosis. So we need to have multiple ways to have an authentication for that kind of critical systems. Hi, I am Solsh from this residential japan And I'm not in the cloud in this biometric field, but I have a question that I need to £500 for it. and uh The first question is you only use it's a fingerprint data to produce um signature but i mean signing key Or do you use other information as I put together with fingerprint. And second question is. do you have any um published the paper on the algorithm on generating Saniki from the fingerprint Because I'm not sure if you can maintains a balance between the duplicating of the sariki And it will be so thank you. About first question, it depends on It depends on biometric signature scheme For example, positive signature profile from Hitachi uses only fingerprint. or fisher ditto and And for example. For another example, initially also proposed biometric signature scheme and In this signature scheme, we use spametric information and another HEPA data generated by the generated by information But… This data is But we consider this HEPA data is not confidential information because we cannot recover biology of information from that

HERPA data. Sorry, ETO. I'm getting to the 18th. get the pot. Thank you. Eric just put it in the chat that liveness is an old issue that's been fairly well addressed recently but uh Okay. Generative AI is going to make it uh a challenge in the future. The best thing about biometrics is I've done well It works for you and only you. The problem with biometrics is It represents you and only you. If you lose, because you might be thinking, oh, you know, I've got a a ledger nano. And if I lose the Ledger Nano, I've lost my wallet. Well, if you lose your finger, you've lost your wallet. You know, it's a much bigger system than just capturing a fingerprint or a face or a scan or an iris. There are all sorts of issues of… Literally key recovery. Which is now a lot harder because, you know, someone is missing parts and people do lose body parts. And if there is a compromise, there was a question about, you know, like, is this a JPEG that's easily copied? The answer is not for about 30 years. You can imagine, again, probably with generative AI, being able to reverse engineer the, and you can think of it conceptually as data point hashes. You shouldn't be able to reconstruct something. I would not put it p

Could I quickly do my section of the presentation and then we move on to wallet governance? So I'll make this fairly quick because it's more context setting than the specific research that we've conducted at the IKP Working Group. And it's also considered, I think this part of the conversation is Also a bit of a thought-provoking call to action around the questions of wallet governance. uh and a way for us to build that, begin motivation of shared common understanding and language. When talking about wallets. And just to set a little bit of context for this. is when we started speaking about, well, it's, I guess, in the IKP working group we started to have a lot of different opinions on what is a wallet. And then we started to think about well. There are identity wallets, there are biometric wallets, there are wallets that deal with finance. There is a leather wallet in my pocket that used to hold more money than it does today um because most of it is digital, right? So how do we build a shared understanding around wallets? And this is the nature of the paper that we're preparing in Serkisan has contributed a lot of the fundamental research behind the sort of wallet typologies i guess So yeah, some of the key questions around this then lead to governance decisions about which wallets are best used and almost like best practice for different use cases in the ecosystem. So when we're talking about treasury management. what type of wallet should we use? Should we use a combination? We've got metric bound wallets as well as a multi-seq or NPC solution? And this is really what we're trying to interrogate in this paper. to try and provide guidance to not only the industry, but also like the users of wallets at the end of the day as to what services they access and different wallets that they use for different purposes. And then just as a to like push shadow the end of this part We then start to ask the fundamental question of there is a convergence between the identity side of wallets and the financial use case of wallets. And at one point in the future, we're going to be using the same wallet for multiple purposes. So should we actually have this separation at all? So the paper. goes through three main categories. One being the wallets for financial use cases Well, it's the governance use cases and while it's your identity. In the last couple working the later two meetups. that we presented this, we actually had some really interesting feedback where Some people in the audience said that wallets for governance is kind of like a mix of both. So should it

even be a section in of itself I still stand on the opinion that wallets for governance is going to be an emerging space where particular signing keys and wallets are fit for purpose and only used for governance decisions. However, I think that, yeah, that is up for debate and would like to continue talking about that in the working groups and today. So that's where we'll get to the call for participation So I understand there's a lot of words on these slides. I won't read it to you, but kind of the idea and it's a little bit of insight into what's been written in the paper. We go through these sort of primary use cases and almost like the best practice. So when we talk about custodial wallets. Where are they used in the cryptocurrency space? There's a lot more sort of custody providers emerging in the space as well. So what type of underlying technology do they use? for managing custodial wallets. And because it's non-custer wallets, hardware wallets multi-seq, and then more recently smart contract wallets, which I think are the emerging space where we can have this uh sort of layered approach to the usage of wallets. So a smart contract wallet allows for me to build something called account abstraction Which means I can use a different key for different purposes. And manage that. So some of the questions when we talk about the financial use cases of wallets. non-financial and financial use cases merging? Should they be merged? Should we use different signing keys and different wallets for these things? One of the questions that I'm focusing on more recently is wallets for AI agents. AI agents are going to be using blockchain-based wallets. Should there be some sort of signature or tag in the ecosystem saying that this wallet is controlled by AI as opposed to this wallet is a human who's sort of managing the risk and then delegating control to an AI agent. And then the final question around finance is the reputation systems as they slowly move on chain. Does this affect that sort of unification of the two? Of course, I spoke a little bit about wallets for governance. And I think the accountable wallet research that Masato has done is also very much related to wallets for governance. But they get used for voting in sort of on-chain governance. All of the time now. There's different types of dog governance, there's staking protocol, DAO-based governance, transactional, so like the rules-based system, the capital-based firm would go through in order to commit a transaction send money via OTC, for example. how do we build these systems into best practice is something else that's outlined in the study report. One interesting question that I've come up with is, are there any types of wallets that we shouldn't use? for governance. So should we not allow, for example, hot wallets to be used for significant governance decisions. when like interacting with the DAO. in order to esteem best practice of key management. And then finally, wallets for identity. Here we have this evolution over time. Typically, it's kind of broken up into the verifiable credential space, the decentralized identity potentially uh sorry that did or decentralized identifier space And then, of course, we think about the self-sovereign identity principles that can be built into the wallet-based systems as well for identity. So I just wanted to sort of go through that really quickly to add context to the paper that we've been working on with wallet governance that has contributed to with the biometric piece, as well as plenty of other sort of classifications of the types of wallets and then bring up this discussion point for the room. around this convergence of a one size fits all or a wallet to rule them all in a way um so I think that these are really interesting questions.

I will sort of move the conversation into Masato's presentation. So he has ample time to explain the accountable wallet research and subject that we've been studying, analyzing, and preparing. But just keep those questions in mind for the discussion afterward.

Let me share my screen. Oops.

Today, I have to give an overview of the accountable wallet concept, which is an ongoing discussion topic within IKP, our working group of Bing. Since the session time is very limited, I will keep my presentation short and won't dive into the technical discussion today, focusing on the idea itself.

We define the accountable wallet as a wallet that can prove your legitimacy to others. I'm going to explain the "how" about later, so let me begin by explaining the research background. First, accountable wallets were not created for money laundering, obviously. However, in reality, they are being used for it. Agencies like the FBI have pointed out that so-called coin mixers, especially Tornado Cash, are used for money laundering.

Money laundering here isn't limited to hacking proceeds; it can include anything from illegal drug sales proceeds. These proceeds, if not yet in crypto, can be converted from fiat to crypto through anonymous transactions like TOR or P2P transactions. However, if the launderer sends their crypto to exchanges directly from illegal platforms, it can be detected as an illegal source. So, they typically route it through mixers like Tornado Cash before depositing it into exchanges and then treat it as legitimate investment profits and pay taxes accordingly.

Here's a dilemma: Tornado Cash is officially prohibited in the United States, and exchanges in the US are prohibited from receiving crypto from wallets that used Tornado Cash. However, launderers know this, so instead of depositing directly from a wallet that used Tornado Cash, they make detours through several wallets before depositing into exchanges. Even if exchanges can detect indirect Tornado Cash usage, the recipients may not have used it themselves, perhaps just receiving crypto as payment for legitimate goods or services. From an on-chain perspective, these transactions appear identical to those from Tornado Cash.

In other words, it's impossible to determine from on-chain data alone whether a user is malicious or not, or simply acquired crypto in good faith. Privacy-enhancing tools like Tornado Cash are necessary, as crypto transactions are otherwise easily traceable. However, using Tornado Cash, even unknowingly, can mean you are facilitating laundering.

In most jurisdictions, receiving criminal proceeds unknowingly doesn't result in charges, termed good faith acquisition. However, proving good faith is challenging, as some may feign ignorance after knowingly buying crypto at a slight discount from criminal sources. This brings an inevitable trade-off between privacy and compliance, leaving users without a way to verify crypto sources.

This is why we are developing technologies enabling users to comply with regulations on their own. Currently, it’s difficult for a recipient to verify the sender's legitimacy from on-chain data alone. If legitimacy could be proven via verifiable proofs, the process would be simplified. Our working group is discussing proof requirements, issuance procedures, and privacy protection within the protocol.

Another problem is that the definition of legitimacy varies across jurisdictions. For instance, some agencies publish a sanction list for Bitcoin addresses, but it’s up to each participant to decide whether to transact with wallets on these lists. Even if you check your direct counterparty, it’s hard to verify indirect paths. Launderers hide identities, complicating the detection of prohibited addresses.

This is where compliance incentives come in, which is our concept of the accountable wallet. Last year, co-authors proposed privacy pools, a mixing protocol allowing users to mix assets only among legitimate users, separating them from blacklisted users. This privacy pool concept technically preserves transaction privacy and prevents mixing with laundered funds.

Users receive a membership proof with ZK technology, proving their withdrawal was mixed only among legitimate users. These association sets, defined by whitelists or blacklists, are created beforehand by a trusted party. This allows law-abiding users to prove legitimacy while preserving privacy.

For example, someone could hold legally obtained crypto and mix it within these association sets. Privacy pools thus enable legitimate mixing without laundering. Membership proofs can be presented in subsequent transactions.

To create synergy between privacy pools and accountable wallets, association sets can be joined by users proving legitimacy through accountable wallets. Association sets allow users to prove compliance to a common standard. For instance, some users may prefer to mix only with US residents, forming customized association sets.

Thus, mixing within these sets provides a membership proof, ensuring crypto assets weren’t mixed with illegal wallets. This means illegitimate wallets are defined differently in each set, allowing users to customize legitimacy.

We are currently working on a technical document for implementing these proofs and discussing incentive mechanisms and privacy considerations. If you're interested in joining this project, we encourage you to get involved.

Thank you for the presentation. I'll open the floor for questions from the audience. If there are remote questions, I can see and read them, or you can unmute.

One of the biggest pieces of feedback from presenting the accountable wallet concept at Bitcoin Tokyo was the duality it creates in the ecosystem—a division between accountable and non-accountable ecosystems. There's a question of whether this could affect token fungibility, like Bitcoin. For example, could an accountable Bitcoin be priced higher than a non-accountable one? This concern over Bitcoin's ecosystem was an interesting topic at Bitcoin Tokyo.

Another question is how to build incentives for using accountable wallets. Positive ecosystem inclusion might itself be enough incentive for users, as people often want to participate in trusted ecosystems. This inclusion could create a snowball effect, making accountable wallets necessary for accessing specific opportunities.

In decentralized finance, compliant, walled gardens are another potential use for accountable wallets. Block filters could verify whether a transaction comes from a sanctioned wallet ecosystem. There's room for reputation-based incentives, business risk mitigation, and benefits for individual investors.

Any more questions or comments? It's the last session of BGIN for the day.

Thank you for the presentations. I have a question about user responsibility. Recognizing privacy rights pushes the responsibility of privacy management onto users, shifting boundaries between providers and users. Expanding privacy comes with added user responsibility, which is ironic. Will accountable wallets ease or increase user burdens in managing privacy?

Thank you for this question. Due to crypto's boundless nature, a unified legitimacy standard is impossible worldwide. We can't force compliance through laws alone. That's why incentive strategies are critical. Offering incentives might encourage people to use accountable wallets.

Similar to real-world credit, users may receive favorable terms if they can prove trustworthiness. Accountable wallets function similarly: proving legitimacy can unlock benefits like lower interest rates or better DeFi terms. This concept parallels operational security and data ownership, as privacy management burdens now fall on users. Accountable wallets offer a chance to shift this dynamic to a positive-sum outcome where users benefit by taking ownership of their data.

Thank you for your comments. We have another question. In the previous session, we discussed risks stakers face when validating illicit transactions. Could accountable wallets benefit validators as well, alongside VASPs?

In other words, users can also benefit from having a transaction validated if they use the accountable wallet.

Yeah, I don't disagree with it. It's worth exploring the possibility, but personally, I'm kind of against it. I've heard that there was an incident where a huge Bitcoin mining pool, F2Pool from

China, intentionally excluded some transactions flagged as sanctions by OFAC. After the exclusion, they admitted to it but decided to stop doing so after criticism, as the decision was very centralized.

This idea of the accountable wallet should form compliance through decentralized, individual decisions. If each person decides to exclude certain addresses, it could cause significant issues for that address. The idea of staking or mining pools feels very centralized, which is why I'm hesitant.

It's interesting. I had a similar thought, and I think we can avoid the problem you outlined. If I had an accountable wallet attestation and was part of an association set allowing transactions only mined by particular relayers that check against OFAC, it could be a useful application of the technology and concept. We'd need to analyze the ramifications, but there's synergy between the technologies. Accountable wallets and privacy pools could be quite complementary, as could NEV bots focused on compliance rather than extracting value.

Thank you.

I just had a question after hearing your point about voluntary-based adoption. I'm not sure if users will adopt this on their own to avoid involvement in harmful activities. A developer in Tokyo once said people don't often use privacy coins because they don't prioritize privacy. Avoiding illegal involvement might be different from privacy protection, but adoption could still be an issue.

I support voluntary-based accountable wallets but wonder if it'll be adopted just because users want to avoid bad actors. Perhaps some enforcement might be needed.

One thing I like about the accountable wallet concept is that it's not just a negative association set. It could also work positively, like a whitelist for accessing airdrops or new token opportunities. Over time, it could build an on-chain reputation system, encouraging financial incentives for inclusion. Eventually, this could ease acceptance of illegitimate association sets and attestations.

I'm not a developer, but I believe this would need a serious approach. Using it should help protect users from negative interactions. But there must be consequences for non-compliance, even if users don't have to interact with others who aren't compliant.

From my understanding, usage would involve signing an additional transaction or querying an on-chain registry for membership proof, similar to connecting a wallet to a protocol. Adding an extra signature step may be more complex on the development side, but it could be worth it.

Once someone creates an accountable wallet, can it be revoked in the future? Would transactions be monitored, and if so, does this raise privacy concerns?

Good question. I don’t think there must be monitoring for this. Revocation could be isolated on each occasion, with legitimacy decaying over time. Closer transactions to the block height would have higher accountability. Wallet holders can refresh their legitimacy if they maintain control. If a bad actor takes over, the trusted third party could revoke the attestation.

Thank you.

I think your idea about tying it to association sets is helpful. My question was more about the steady state of an accountable wallet itself. Bad actors might initially appear clean, but how would monitoring work over time? Could a wallet’s accountability be revoked if it’s later misused?

A good point. There would need to be real-time decay in the system. If I maintain control, I can refresh my reputation. However, if a bad actor gains control, the third party would recognize it and revoke the token, possibly affecting related attributions.